Project 1 Progress: Machine Learning-Based ET0 Estimation

Objective: Estimate daily reference evapotranspiration (ET0) for agricultural irrigation using limited weather variables.

5,113 Days	14 Years	0 Missing Days	100% Match
Daily Observations	2010 - 2023 Period	Clean Verified Data	Exact User CSV Audit

1. Work Completed So Far (Weeks 1 & 2 Deliverables)

[Step 1 Completed] Meteorological Data Acquisition & Verification (NASA POWER API)

Fetched 14 years (2010 - 2023) of daily agroclimatology data for **Hisar, Haryana** (Lat: 29.15 N, Lon: 75.72 E, Elev: 215m) matching the base research paper (Shaloo et al.). Verified 0 missing values, checked physical bounds, and audited against the manual user download with a **100.0% row-by-row exact match**.

[Step 2 Completed] Ground-Truth ET0 Formulation (FAO-56 Penman-Monteith)

Implemented the complete standardized FAO-56 Penman-Monteith physical equation to compute reference evapotranspiration (ET0 in mm/day). The target variable averages **5.14 mm/day** (0.65 mm in winter to 14.5 mm in peak summer). Verified against Python's certified pyet library with an absolute discrepancy under 0.00012 mm/day.

[Step 3 Completed] Self-Contained Kaggle / Jupyter Notebook Created

Created evapotranspiration_fao56_calculation.ipynb. It features automatic data detection for Kaggle input and local folders, handles NASA metadata headers, runs without external dependencies, generates EDA time-series and monthly boxplots, and exports the clean dataset ready for model training.

2. Summary of Weather Variables in Dataset

Variable Name	Dataset Column	Unit	Mean +/- Std	Range	Role in Machine Learning
Max Temperature	max_temp	deg C	33.15 +/- 7.20	13.4 - 47.9	Input Feature (All 3 Configurations)
Min Temperature	min_temp	deg C	19.49 +/- 8.18	0.2 - 34.7	Input Feature (All 3 Configurations)
Relative Humidity	relative_humidity	%	49.20 +/- 18.23	11.4 - 94.7	Input Feature (Configurations 2 & 3)
Wind Speed (2m)	wind_speed	m/s	2.42 +/- 0.88	0.6 - 7.9	Input Feature (Configuration 3 only)
Solar Radiation	solar_radiation	MJ/m2/d	17.52 +/- 5.69	1.4 - 29.9	Used to calculate ground-truth ET0 target
Surface Pressure	surface_pressure	kPa	98.81 +/- 0.54	96.9 - 100.3	Atmospheric pressure for psychrometric constant
Target ET0	**ET0**	**mm/day**	**5.14 +/- 2.37**	**0.65 - 14.53**	**Prediction Target (y) for all ML models**

3. Next Phase: Machine Learning Configurations to Evaluate

Config 1: Temperature Only	Config 2: Temp + Humidity	Config 3: Temp + Humidity + Wind
Features: max_temp, min_temp Simulates the cheapest agricultural weather station ($5 sensor).	Features: max_temp, min_temp, RH Captures air drying power and vapor pressure deficit.	Features: max_temp, min_temp, RH, WS Includes aerodynamic wind transport of water vapor.

Core Research Question: *Can machine learning estimate ET0 accurately using only temperature and humidity, allowing farmers to eliminate expensive solar radiation sensors?*

Page 2: Student Questions for Project Mentor

3 practical queries from our student team regarding model training, data splitting, and demonstration requirements.

Question 1: Model Training Guidance

We haven't learned how to train machine learning models yet in class - how should our team approach training this?
Could you guide us on the starter setup? Is it appropriate for us to use Python's scikit-learn library with standard baseline regressors (Linear Regression, Random Forest Regressor, and SVR) to train on our weather inputs?

Question 2: Dataset Train / Test Split Ratio

In model training, should our dataset split be 80/20 or 70/20/10?
We have 14 years of daily data (5,113 days). Should we use an **80% Train / 20% Test** split, or does the course grading rubric require a 3-way **70% Train / 20% Validation / 10% Test** split?

Question 3: End-Semester Demonstration Timeline

Can we expect to finish this for a complete working demonstration before end-sem?
What are the specific evaluation deliverables expected from our team during the final project demonstration (e.g. running the live Kaggle notebook, presenting the comparison table, and showing Predicted vs. Actual scatter plots)?

Target Evaluation Metrics for Model Comparison

MAE (Mean Absolute Error)	RMSE (Root Mean Squared Error)	R2 (Coefficient of Determination)
Measures average magnitude of prediction error in mm/day. Target benchmark: < 0.6 mm/day.	Penalizes large errors heavily; important during heatwave spikes. Target: < 0.9 mm/day.	Percentage of daily ET0 variance explained by the model. Target benchmark: > 0.90.

Roadmap to Complete Demonstration Before End-Sem

Step 1: Train the 3 Models (Week 3): Implement Linear Regression, Random Forest, and SVR across the 3 weather configurations (9 total runs).

Step 2: Build Evaluation Table (Week 3): Compile test scores for MAE, RMSE, and R2 into a side-by-side comparison matrix.

Step 3: Generate Visual Plots (Week 4): Produce Predicted vs. Actual scatter plots and Random Forest feature importance rankings.

Step 4: Final Live Demo & Viva (Before End-Sem): Demonstrate the complete Kaggle notebook and answer the core research question.